

Stat 6841 Homework 4

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Problem 1.

We need to show that $E[|M_n|] < \infty$ for $n \geq 1$, that $\{M_n\}$ is adapted to $\{\mathcal{F}_n\}$, and that $E[M_{n+1}|\mathcal{F}_n] = M_n$ for $n \geq 1$.

1. We have

$$\begin{aligned} E[|M_n|] &= E[|S_n^2 - v_n|] = E\left[\left|\left(S_0 + \sum_{i=1}^n \xi_i\right)^2 - \sum_{i=1}^n \sigma_i^2\right|\right] \\ &\leq E\left[\left|S_0 + \sum_{i=1}^n \xi_i\right|\right] + \sum_{i=1}^n \sigma_i^2 \\ &\leq S_0^2 + \sum_{i=1}^n E[\xi_i^2] + \sum_{i=1}^n \sigma_i^2 < \infty. \end{aligned}$$

2. Since M_n is a continuous function of $\xi_1, \dots, \xi_n$, and ξ_i is $\mathcal{F}_n$ -predictable for $i = 1, \dots, n$, it follows that M_n is $\mathcal{F}_n$ -predictable. Thus, $\{M_n\}$ is adapted to $\{\mathcal{F}_n\}$.

3. We have

$$\begin{aligned} E[M_{n+1}|\mathcal{F}_n] &= E\left[\left(S_0 + \sum_{i=1}^{n+1} \xi_i\right)^2 - \sum_{i=1}^{n+1} \sigma_i^2 \middle| \mathcal{F}_n\right] \\ &= E\left[\left(S_0 + \sum_{i=1}^n \xi_i\right)^2 + 2\left(S_0 + \sum_{i=1}^n \xi_i\right)\xi_{n+1} + \xi_{n+1}^2 - \sum_{i=1}^{n+1} \sigma_i^2 \middle| \mathcal{F}_n\right] \end{aligned}$$

Since $\left(S_0 + \sum_{i=1}^n \xi_i\right)^2$ is a continuous function of $\xi_1, \dots, \xi_n$, it is $\mathcal{F}_n$ -predictable because $\xi_1, \dots, \xi_n$ are all $\mathcal{F}_n$ -predictable. Then

$$E[M_{n+1}|\mathcal{F}_n] = \left(S_0 + \sum_{i=1}^n \xi_i\right)^2 - \sum_{i=1}^n \sigma_i^2 + 2\left(S_0 + \sum_{i=1}^n \xi_i\right)E[\xi_{n+1}|\mathcal{F}_n] + E[\xi_{n+1}^2|\mathcal{F}_n] - \sigma_{n+1}^2.$$

Since ξ_{n+1} is independent of $\mathcal{F}_n$, it follows that ξ_{n+1}^2 is also independent of $\mathcal{F}_n$, and $E[\xi_{n+1}|\mathcal{F}_n] = E[\xi_{n+1}] = 0$, and $E[\xi_{n+1}^2|\mathcal{F}_n] = E[\xi_{n+1}^2] = \sigma_{n+1}^2$. Therefore,

$$E[M_{n+1}|\mathcal{F}_n] = \left(S_0 + \sum_{i=1}^n \xi_i\right)^2 - \sum_{i=1}^n \sigma_i^2 = M_n.$$

Problem 2.

- (a) Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Then $\{M_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$. To show this, we need to show that $E[|M_n|] < \infty$ for $n \geq 1$, that $\{M_n\}$ is adapted to $\{\mathcal{F}_n\}$, and that $E[M_{n+1}|\mathcal{F}_n] = M_n$.

1. We have

$$E[|M_n|] = E[2^n X_n] = 2^n E[U_1 \cdots U_n] = 2^n \prod_{i=1}^n E[U_i] = 1 < \infty$$

because $U_1, \dots, U_n$ are independent with $E[U_i] = \frac{1}{2}$ for $i = 1, 2, \dots, n$.

2. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Then M_n is a continuous function of $X_1, \dots, X_n$, each of which is $\mathcal{F}_n$ -predictable, so M_n is $\mathcal{F}_n$ -predictable.

3. We have

$$E[M_{n+1}|\mathcal{F}_n] = E[2^{n+1}X_{n+1}|\mathcal{F}_n] = 2^{n+1}E[X_n U_n|\mathcal{F}_n].$$

Since U_n is independent of $\mathcal{F}_n$ and X_n is $\mathcal{F}_n$ -predictable, we have

$$E[M_{n+1}|\mathcal{F}_n] = 2^{n+1}X_n E[U_n|\mathcal{F}_n] = 2^{n+1}X_n E[U_n] = 2^n X_n = M_n.$$

- (b) Note that $-\log(U_i)$ is exponentially-distributed with mean 1 because, by the CDF method,

$$P(-\log(U_i) \leq x) = P(U_i \leq e^{-x}) = e^{-x}, \quad x \geq 0.$$

Then $E[\log(U_i)] = -E[-\log(U_i)] = -1$, and, by the Strong Law of Large Numbers,

$$\frac{1}{n} \log(X_n) = \frac{1}{n} \sum_{i=1}^n \log(U_i) \xrightarrow{\text{a.s.}} -1$$

as $n \rightarrow \infty$.

- (c) To show that $M_n \rightarrow 0$ almost surely, we must show that $P(B) = 1$, where $B = \{M_n \rightarrow 0\}$. Since $P(B) \leq 1$ in any case, we only need to show that $P(B) \geq 1$.

To this end, let $A = \{\frac{1}{n}M_n \rightarrow \log(2) - 1\}$. Then $P(A) = 1$ because, by (b),

$$\frac{1}{n} \log(M_n) = \log(2) + \frac{1}{n} X_n \xrightarrow{\text{a.s.}} \log(2) - 1.$$

The natural exponent base e is greater than 2, so $\log(2) < 1$, and there exists $\delta > 0$ such that $\log(2) + \delta < 1$. Let $\omega \in A$, and let $\varepsilon > 0$. Since $\omega \in A$, there exists $N_1 > 0$ such that

$$\frac{1}{n} \log(X_n(\omega)) < -\log(2) - \delta \quad \text{for all } n > N_1.$$

Choose $N > \max\left\{N_1, -\frac{\log(\varepsilon)}{\delta}\right\}$. Then for all $n > N$, we have

$$\begin{aligned} \frac{1}{n} \log(X_n(\omega)) < -\log(2) - \delta &\implies \log(X_n(\omega)) + n \log(2) < -n\delta \\ &\implies |M_n(\omega)| < e^{-n\delta} < e^{-N\delta} < \varepsilon. \end{aligned}$$

Thus, $M_n(\omega) \rightarrow 0$ as $n \rightarrow \infty$, so $\omega \in B$. Since $\omega \in A$ was arbitrary, it follows that $A \subseteq B$, and $1 = P(A) \leq P(B)$. This proves that $M_n \rightarrow 0$ almost surely.

Problem 3.

T_1 and $T_1 + 1$ are stopping times, but $T_1 - 1$ is not.

1. T_1 is a stopping time because

$$\{T_1 \leq n\} = \bigcup_{i=1}^n \{M_i = 1\} \in \mathcal{F}_n$$

because $\{M_i = 1\} \in \mathcal{F}_i \subseteq \mathcal{F}_n$ for $i = 1, 2, \dots, n$.

2. $T_1 + 1$ is a stopping time because

$$\{T_1 + 1 \leq n\} = \{T_1 \leq n - 1\} = \bigcup_{i=1}^{n-1} \{M_i = 1\} \in \mathcal{F}_n$$

because $\{M_i = 1\} \in \mathcal{F}_i \subseteq \mathcal{F}_n$ for $i = 1, 2, \dots, n - 1$.

3. $T_1 - 1$ is not necessarily a stopping time because

$$\{T_1 - 1 \leq n\} = \{T_1 \leq n + 1\} = \{M_{n+1} = 1\} \cup \bigcup_{i=1}^n \{M_i = 1\} = \{M_{n+1} = 1\} \cup \{T_1 \leq n\}.$$

Since $\{M_{n+1} = 1\}$ may not be in $\mathcal{F}_n$, we cannot conclude that $\{T_1 - 1 \leq n\} \in \mathcal{F}_n$.

Problem 4.

Let A_s be the event that $X_t = -1$ for all $t > s$, for $s = 1, 2, \dots$, that is,

$$A_s = \bigcap_{t>s} \{X_t = -1\}.$$

For an outcome $\omega \in A_s$, it is clear that $M_t(\omega) \rightarrow -\infty$. Thus, $A_s \subseteq \{M_t \rightarrow -\infty\}$ for all $s = 1, 2, \dots$. Then $\{M_t \rightarrow -\infty\}^C \subseteq A_s^C$ for all $s = 1, 2, \dots$.

By Boole's inequality, it follows that

$$P(A_s^C) = P\left(\bigcup_{t>s} \{X_t = t^2 - 1\}\right) \leq \sum_{t>s} P(X_t = t^2 - 1) = \sum_{t>s} \frac{1}{t^2}.$$

Then $P(A_s^C) \rightarrow 0$ as $s \rightarrow \infty$ because the last expression above is the tail of the convergent sum $\sum_{t=1}^{\infty} \frac{1}{t^2}$.

Hence, by the squeeze theorem,

$$0 \leq P(M_t \not\rightarrow -\infty) \leq P(A_s^C) \implies P(M_t \not\rightarrow -\infty) = 0.$$

This means that $P(M_t \rightarrow -\infty) = 1$, that is, $M_t \rightarrow -\infty$ almost surely.

Problem 5.

Since $\{X_t\}$ is a sub-martingale, there is a filtration $\{\mathcal{F}_t\}$ to which $\{X_t\}$ is adapted. We can write

$$X_{t \wedge \tau} = I(\tau \geq t)X_t + \sum_{s=0}^{t-1} I(\tau = s)X_s.$$

Since τ is a stopping time, $I(\tau = s)$ is $\mathcal{F}_s$ -predictable and therefore $\mathcal{F}_t$ -predictable, and $I(\tau > t)$ is $\mathcal{F}_t$ -predictable. Furthermore, X_t is $\mathcal{F}_t$ -predictable, and X_s is $\mathcal{F}_t$ -predictable for $s \leq t$. Since $X_{\tau \wedge \tau}$ is a continuous combination of these variables, it follows that $X_{t \wedge \tau}$ is $\mathcal{F}_t$ -predictable. Thus, $\{X_{t \wedge \tau}\}$ is adapted to $\{\mathcal{F}_t\}$.

Next, we note that

$$E[|X_{t \wedge \tau}|] \leq E[I(\tau \geq t)|X_t|] + \sum_{s=0}^{t-1} E[I(\tau = s)|X_s|] \leq E[|X_t|] + \sum_{s=0}^{t-1} E[|X_s|] < \infty$$

because $\{X_t\}$ is a submartingale.

Finally, we have

$$E[X_{(t+1) \wedge \tau} | \mathcal{F}_t] = E[I(\tau \geq t+1)X_{t+1} | \mathcal{F}_t] + \sum_{s=0}^t E[I(\tau = s)X_s | \mathcal{F}_t].$$

Since $\{\tau \geq t+1\} = \{\tau \leq t\}^C$, and τ is a stopping time, it follows that $\{\tau \geq t+1\} \in \mathcal{F}_t$. Furthermore, $I(\tau = s)X_s$ is $\mathcal{F}_t$ -predictable for $s = 0, 1, \dots, t$. Then

$$\begin{aligned} E[X_{(t+1) \wedge \tau}] &= I(\tau \geq t+1)E[X_{t+1} | \mathcal{F}_t] + \sum_{s=0}^t I(\tau = s)X_s \\ &\geq I(\tau \geq t+1)X_t + I(\tau = t)X_t + \sum_{s=0}^{t-1} I(\tau = s)X_s \\ &= I(\tau \geq t)X_t + \sum_{s=0}^{t-1} I(\tau = s)X_s \\ &= X_{t \wedge \tau}, \end{aligned}$$

so $\{X_{t \wedge \tau}\}$ is a submartingale.

Problem 6.

To show that $\{Y_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$, where $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, we need to show that $E[|Y_n|] < \infty$ for all n , that $\{Y_n\}$ is adapted to $\{\mathcal{F}_n\}$, and that $E[Y_{n+1} | \mathcal{F}_n] = Y_n$.

1. We have

$$E[|Y_n|] = E\left[\left|S_n^2 - \sum_{k=1}^n X_k^2\right|\right] \leq E\left[\left(\sum_{k=1}^n X_k\right)^2\right] + \sum_{k=1}^n E[X_k^2] \leq E\left[n^2 \sum_{k=1}^n X_k^2\right] + n\sigma^2 \leq (n^3 + n)\sigma^2 < \infty.$$

because for any set of real numbers $\{a_i : i = 1, 2, \dots, n\}$, we have

$$\left(\sum_{i=1}^n a_i\right)^2 \leq \left(\sum_{i=1}^n |a_i|\right)^2 \leq \left(n \max_{i=1,2,\dots,n} |a_i|\right)^2 = n^2 \max_{i=1,2,\dots,n} a_i^2 \leq n^2 \sum_{i=1}^n a_i^2.$$

2. It is easy to see that Y_n is a continuous function of $X_1, X_2, \dots, X_n$, so Y_n is adapted to $\mathcal{F}_n = \sigma(X_1, X_2, \dots, X_n)$ for all n .

3. Since X_{n+1} is independent of $X_1, \dots, X_n$, it follows that X_{n+1} is independent of $\mathcal{F}_n$. Thus,

$$\begin{aligned}
 E[Y_{n+1}|\mathcal{F}_n] &= E \left[\left(\sum_{k=1}^{n+1} X_k \right)^2 - \sum_{k=1}^{n+1} X_k^2 \middle| \mathcal{F}_n \right] \\
 &= E \left[\left(\sum_{k=1}^n X_k \right)^2 + 2X_{n+1} \sum_{k=1}^n X_k + X_{n+1}^2 - \sum_{k=1}^n X_k^2 - X_{n+1}^2 \middle| \mathcal{F}_n \right] \\
 &= \left(\sum_{k=1}^n X_k \right)^2 - \sum_{k=1}^n X_k^2 + 2E[X_{n+1}|\mathcal{F}_n] \sum_{k=1}^n X_k \\
 &= Y_n + 2E[X_{n+1}] \sum_{k=1}^n X_k \\
 &= Y_n.
 \end{aligned}$$

We have $S_n = \sum_{k=1}^n X_k^2$. Since $\{X_k\}$ are independent, so, too, are $\{X_k^2\}$. Furthermore, we have $E[X_k^2] = \sigma^2$ for all k . Then, by Wald's Equation,

$$E[S_\tau] = E[X_1^2]E[\tau] = \sigma^2 E[\tau].$$

Problem 7.

Let $Y_n = e^{2b(S_n - bn)}$. Then $\{Y_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$, where $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. To show this, we need to prove that $E[|Y_n|] < \infty$ for all n , that $\{Y_n\}$ is adapted to $\{\mathcal{F}_n\}$, and that $E[Y_{n+1}|\mathcal{F}_n] = Y_n$.

1. Since X_k is standard normal for $k = 1, 2, \dots$, and $S_n = X_1 + X_2 + \dots + X_n$, it follows that S_n is also normal with mean 0 and variance n , so that the PDF of S_n is

$$f(s) = \frac{1}{\sqrt{2\pi n}} e^{-\frac{1}{2n}s^2}.$$

Then

$$E[|Y_n|] = E \left[e^{2b(S_n - bn)} \right] = \int_{-\infty}^{\infty} e^{2bs - 2bn} \frac{1}{\sqrt{2\pi n}} e^{-\frac{1}{2n}s^2} ds \lesssim \int_{-\infty}^{\infty} e^{-\frac{1}{2n}(s - 2nb)^2} ds < \infty.$$

2. Y_n is a continuous function of $X_1, \dots, X_n$, so Y_n is adapted to $\mathcal{F}_n$ for all n .

3. Since X_{n+1} is independent of $X_1, \dots, X_n$, it is also independent of $\mathcal{F}_n$. Then

$$\begin{aligned}
 E[Y_{n+1}|\mathcal{F}_n] &= E\left[e^{2b(S_{n+1}-b(n+1))}|\mathcal{F}_n\right] = E\left[e^{2b(S_n+X_{n+1}-bn-b)}|\mathcal{F}_n\right] \\
 &= E\left[e^{2b(S_n-bn)}e^{2b(X_{n+1}-b)}|\mathcal{F}_n\right] \\
 &= Y_n E\left[e^{2b(X_{n+1}-b)}\right] \\
 &= Y_n e^{-2b^2} \int_{-\infty}^{\infty} e^{2bx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\
 &= Y_n e^{-2b^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-2b)^2+2b^2} dx \\
 &= Y_n \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\
 &= Y_n,
 \end{aligned}$$

where the second-to-last line follows from the change of variables $u = x - 2b$, and the last line follows from the fact that the integrand is the PDF of a standard normal random variable.

All martingales are submartingales, so $\{Y_n\}$ is also a submartingale. Additionally, $Y_n > 0$ for all n . Hence, by the Doob submartingale inequality,

$$P(Y_n > e^{2bc}) \leq P\left(\max_{1 \leq k \leq n} Y_k > e^{2bc}\right) \leq e^{-2bc} E[Y_n].$$

Since Y_n is a martingale,

$$\begin{aligned}
 E[Y_n] &= E[Y_1] = E\left[e^{2b(X_1-b)}\right] = e^{-2b^2} E\left[e^{2bX_1}\right] \\
 &= e^{-2b^2} \int_{-\infty}^{\infty} e^{2bx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\
 &= e^{-2b^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-2b)^2+2b^2} dx \\
 &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\
 &= 1.
 \end{aligned}$$

Furthermore,

$$\{Y_n = e^{2bc}\} = \left\{e^{2b(S_n-bn)} > e^{2bc}\right\} = \{2b(S_n - bn) > 2bc\} = \{S_n > bn + c\};$$

therefore,

$$P(S_n > bn + c) \leq e^{-2bc}.$$

Problem 8.

The variance of X_i is given by $\text{Var}[|X_i|] = E[|X_i|^2] - E[|X_i|]^2$. Since $\text{Var}[X_i] \geq 0$, it follows that $E[|X_i|]^2 \leq E[|X_i|^2] = E[X_i^2] = 1$, so $0 \leq E[|X_i|] \leq 1$.

To show that $\{Y_n\}$ is a supermartingale, we first note that Y_n is a continuous function of $X_1, \dots, X_n$, so Y_n is $\mathcal{F}_n$ -predictable. Second, because $\{X_1, \dots, X_n\}$ are independent, we have

$$E[|Y_n|] = E\left[\prod_{i=1}^n |X_i|\right] = \prod_{i=1}^n E[|X_i|] \leq 1 < \infty.$$

Lastly, we have

$$E[Y_{n+1}|\mathcal{F}_n] = E\left[|X_{n+1}|\left(\prod_{i=1}^n |X_i|\right)\middle|\mathcal{F}_n\right] = \left(\prod_{i=1}^n |X_i|\right) E[|X_{n+1}||\mathcal{F}_n] = Y_n E[|X_{n+1}|] \leq Y_n$$

because X_{n+1} is independent of $\mathcal{F}_n$, and $Y_n \geq 0$, and $0 \leq E[|X_{n+1}|] \leq 1$. Hence, $\{Y_n\}$ is a supermartingale.

By the supermartingale inequality with $p = 2$, we have

$$E\left[\left(\max_{1 \leq k \leq n} Y_k\right)^2\right] \leq 2E[Y_0^2] = 2.$$

Since Y_k is also L^1 for all k , and $\text{Var}\left[\max_{1 \leq k \leq n} Y_k\right] = E\left[\left(\max_{1 \leq k \leq n} Y_k\right)^2\right] - E\left[\max_{1 \leq k \leq n} Y_k\right]^2 \geq 0$, we obtain

$$E\left[\max_{1 \leq k \leq n} Y_k\right]^2 \leq E\left[\left(\max_{1 \leq k \leq n} Y_k\right)^2\right] \leq 2,$$

from which the desired result follows.

Problem 9.

Let τ be the first time that X_t exits (a, b) , that is,

$$\tau = \inf\{t \mid X_t \leq a \text{ or } X_t \geq b\}.$$

Let $\omega \in \{\tau \leq t\}$. Since $X_s(\omega)$ is continuous in s and $\tau(\omega) \leq t$, there exists $s \in [0, t]$ such that $X_s(\omega) = a$ or $X_s(\omega) = b$. This implies that

$$\sup_{s \in [0, t]} X_s(\omega) \geq b \quad \text{or} \quad \inf_{s \in [0, t]} X_s(\omega) \leq a.$$

On the other hand, if for some $t \geq 0$,

$$\sup_{s \in [0, t]} X_s(\omega) \geq b \quad \text{or} \quad \inf_{s \in [0, t]} X_s(\omega) \leq a,$$

for some ω , then, by definition, $\tau(\omega) \leq t$, so $\omega \in \{\tau \leq t\}$. This shows that

$$\{\tau \leq t\} = \left\{ \sup_{s \in [0, t]} X_s \geq b \right\} \cup \left\{ \inf_{s \in [0, t]} X_s \leq a \right\}.$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$ and the continuity of X_s in s , it follows that

$$\sup_{s \in [0, t]} X_s = \sup_{s \in [0, t] \cap \mathbb{Q}} X_s, \quad \inf_{s \in [0, t]} X_s = \inf_{s \in [0, t] \cap \mathbb{Q}} X_s.$$

Therefore,

$$\{\tau \leq t\} = \left\{ \sup_{s \in [0, t] \cap \mathbb{Q}} X_s \geq b \right\} \cup \left\{ \inf_{s \in [0, t] \cap \mathbb{Q}} X_s \leq a \right\}.$$

Since $[0, t] \cap \mathbb{Q}$ is countable, and X_s is $\mathcal{F}_t$ -predictable for all $s \in [0, t] \cap \mathbb{Q}$, it follows that the supremum and infimum of X_s over $s \in [0, t] \cap \mathbb{Q}$ are also $\mathcal{F}_t$ -predictable. Then $\{\tau \leq t\} \in \mathcal{F}_t$, so τ is a stopping time.

Problem 10.

Define the random variable τ by

$$\tau = \min\{0 \leq k \leq n \mid X_k \geq b\} \cup \{n\}.$$

Then τ is a stopping time with respect to $\{\mathcal{F}_k\}$, where $\mathcal{F}_k = \sigma(X_0, X_1, \dots, X_k)$ because

$$\{\tau \leq k\} = \{n < k\} \cup \bigcup_{i=0}^{\min\{n, k\}} \{X_i \geq b\}$$

which implies that $\{\tau \leq k\} \in \mathcal{F}_k$, as $\{X_i \geq b\} \in \mathcal{F}_i \subseteq \mathcal{F}_k$ for $i = 0, 1, \dots, \min\{n, k\}$, and $\{n < k\}$ is either the empty set or the entire sample space and consequently belongs to any σ field.

Let $\omega \in \{\max\{X_0, \dots, X_n\} \geq b\}$. Then $X_k(\omega) \geq b$ for some $k = 0, 1, \dots, n$. Then $\tau(\omega) = k$, so $X_\tau(\omega) = X_k(\omega) \geq b$. Conversely, if $\omega \in \{X_\tau \geq b\}$, then $X_\tau(\omega) \geq b$. Since $\tau \leq n$, it follows that $X_k(\omega) \geq b$ for some $k = 0, 1, \dots, n$. Then $\max\{X_0(\omega), \dots, X_n(\omega)\} \geq b$. This shows that

$$\{\max\{X_0, \dots, X_n\} \geq b\} = \{X_\tau \geq b\}.$$

Define $Y_k = X_k - X_{\tau \wedge k}$. Then $\{Y_k\}$ is a submartingale with respect to $\{\mathcal{F}_k\}$. Indeed, since $\{X_k\}$ and $\{X_{\tau \wedge k}\}$ are both submartingales, their difference is adapted to $\{\mathcal{F}_k\}$, and

$$E[|Y_k|] \leq E[|X_k|] + E[|X_{\tau \wedge k}|] < \infty.$$

Lastly, we note that

$$Y_k = I(\tau \leq k)(X_k - X_\tau) = I(\tau < k)(X_k - X_\tau).$$

Thus,

$$E[Y_{k+1}|\mathcal{F}_k] = E[I(\tau < k+1)(X_{k+1} - X_\tau)|\mathcal{F}_k] = E[I(\tau \leq k)(X_{k+1} - X_\tau)|\mathcal{F}_k].$$

Since $\{\tau \leq k\} \in \mathcal{F}_k$, it follows that $I(\tau \leq k)$ is $\mathcal{F}_k$ -predictable. Furthermore, $I(\tau \leq k)X_\tau$ is also $\mathcal{F}_k$ -predictable because for any Borel set B

$$\{I(\tau \leq k)X_\tau \in B\} = (\{\tau > k\} \cap \{0 \in B\}) \cup (\{\tau \leq k\} \cap X_{\tau \wedge k} \in B) \in \mathcal{F}_k.$$

This follows from the fact that $\{\tau \leq k\} \in \mathcal{F}_k$ and $\{\tau > k\} \in \mathcal{F}_k$ because τ is a stopping time, and $\{0 \in B\} \in \mathcal{F}_k$ because $\{0 \in B\}$ is either empty or the whole sample space, and $X_{\tau \wedge k}$ is also a submartingale by Problem 5, so $X_{\tau \wedge k}$ is $\mathcal{F}_k$ -predictable.

Using this, we get

$$E[Y_{k+1}|\mathcal{F}_k] = I(\tau \leq k)E[X_{k+1}|\mathcal{F}_k] - I(\tau \leq k)X_\tau \geq I(\tau \leq k)(X_k - X_\tau) = Y_k.$$

Hence, $\{Y_k\}$ is a submartingale. This implies that

$$E[X_n] - E[X_\tau] = E[X_n - X_{\tau \wedge n}] = E[Y_n] \geq E[Y_0] = E[X_0 - X_{\tau \wedge 0}] = 0.$$

Thus, $E[X_n] \geq E[X_\tau]$, from which we deduce that

$$P(\max\{X_0, \dots, X_n\} \geq b) \leq \frac{E[X_\tau]}{b} \leq \frac{E[X_n]}{b}.$$

Problem 11.

Listing 1: Code to simulate martingales

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
```



```
4
5 def random_bernoulli():
6     return float(np.random.random() < 0.5)
7
8
9 def random_normal():
10    return np.random.normal()
11
12
13 def simulate(n, random_fn):
14     s = np.empty(n)
15     s[0] = random_fn()
16     for i in range(1, n):
17         s[i] = s[i-1] + random_fn()
18     return s
19
20
21 def plot3_bernoulli():
22     paths = [simulate(50, random_bernoulli) for _ in range(3)]
23     n = np.arange(50)
24     for path in paths:
25         plt.plot(n, path)
26     plt.show()
27
28
29 def plot3_normal():
30     paths = [simulate(50, random_normal) for _ in range(3)]
31     n = np.arange(50)
32     for path in paths:
33         plt.plot(n, path)
34     plt.show()
35
36
37 if __name__ == '__main__':
38     plot3_bernoulli()
39     plot3_normal()
```

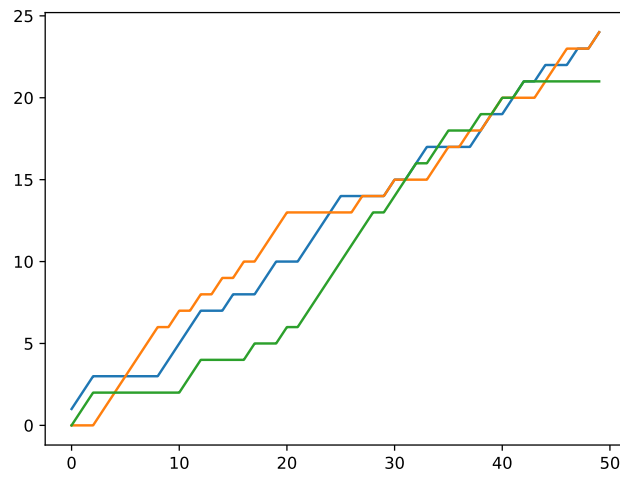


Figure 1: Three paths of $\{S_n\}$ with $X_i \sim B(1, 0.5)$

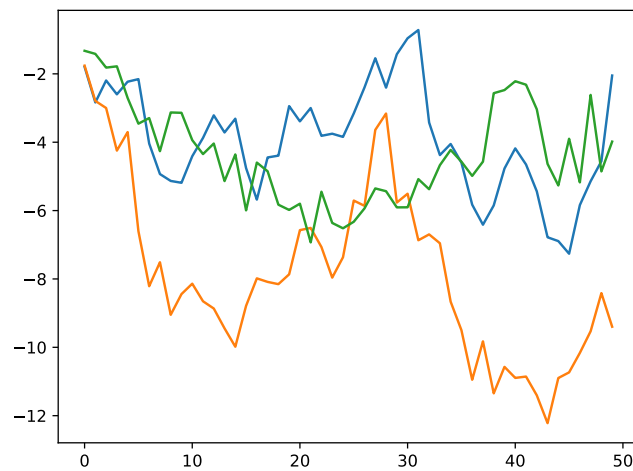


Figure 2: Three paths of $\{S_n\}$ with $X_i \sim N(0, 1)$